CarePlanPlus: Treatment Pathway Recommender System

Harishankar Somasundreams (200043857)
Sowmaya Ramachandappa (200062109)
Kashish Khan (200063919)

Abstract

Although many traditional healthcare recommender systems rely on clinical outcomes, they do not consider patient reported experiences to an extent that supports complete evaluation of healthcare. In this final project we present CarePlanPlus, a new hybrid recommender system supporting clinical outcomes (MMIC-MC) and patient reported experience proxies derived from research national inpatient Experience Survey (NIES). Our recommender system utilizes two complimenting methodological approaches. (1) a similarity role based collaborative and content based filtering model based on matrix factorization and SVM's. This also incorporated patient similarity modeling, as well. And (2) an sequential learning model and approach based on BERT based transformer models, recovery and procedure, details. This categorizing included 309 unique procedures into 12, NIES based condition categorization with 76,388 coverage. The similarity based model provided meaningful recommender systems through similar diagnosis records with satisfaction recommendations ranging from 0 to 1, across multiple medical condition categories using content based filtering for treatment prediction. The sequential model takes advantage of BERT's contextual modeling and sequential learning to predict the procedures of a treatment paths. The model incorporated diagnosis sequences and procedure histories, with patient satisfaction and weighted scoring mechanisms. The incorporation of patient reported satisfaction scores as outcome proxies is a substantial step forward in healthcare personalization, and reduction of barriers between clinical effectiveness and patient experience. We believe that the two approach method will allow more consistent and holistic approaches to healthcare allowing for ultimate possible patient experience.

Keywords: Healthcare Recommender Systems, Patient Similarity, BERT, Sequential Learning, MMICIV, Patient-Reported Outcomes, Collaborative Filtering, Personalized Medicine, Transformer Models

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1 Introduction

Healthcare is increasingly shifting towards personalized medicine which emphasizes treatment tailored to patient-specific characteristics, preferences and experiences (1). Yet, current clinical decision support systems are typically designed to focus on clinical outcomes and medical effectiveness while often downplaying each patient's subjective experience or satisfaction with their care (2). This creates a serious disparity between what clinicians view as successful treatment and what patients view as quality care.

The advent of large-scale healthcare databases such as MMIC-IV (Medical Information Mart for Intensive Care) generates unparalleled capabilities to examine both patterns of care in patients and patient outcomes (3). Simultaneously there has been a emergence of patient-reported outcome measures (PROMs) and patient experience surveys with a specific patient-reported experience survey (NIES), revealing aspects of the patient-perceived value of healthcare-quality (4). The convergence of these various types of data represents a unique opportunity to develop more robust and patient-centric recommendation systems.

The recent rise of natural language processing has been particularly valuable in healthcare, with new transformer models such as BERT (Bidirectional Encoder Representations from Transformers) demonstrating exceptional facility to understand sequential and contextual relationships among healthcare data (5).

1.1 Problem Statement

The issue this project aims to tackle is that healthcare recommender systems focus on clinical outcomes, without capturing patient-reported experiences. By focusing on clinical outcomes, we are missing a disconnecting factor that makes dissonance between treatments that are clinically effective and those that create satisfaction and preference with patients, which may lead to a reduction in the care quality they receive. The health-care system historically thinks in terms of metrics like lab results, and recovery time rather than patient feedback on the experience of care. This is the gap we are advocating for solutions to close. Focusing on clinical outcomes is necessary, but dissonance occurs when making recommendations that might be medically sound but do not resonate with patients. This is particularly problematic if we want patients to be engaged in a plan of care that they will be satisfied with.

1.2 Research Aim

The aim of CarePlanPlus is to develop a hybrid recommender system that provides personalized medical procedure recommendations by integrating clinical data from MIMICIV with patient-reported satisfaction metrics from NIES, thereby improving clinical outcomes and enhancing patient experiences.

1.3 Research Objectives

- Develop a comprehensive personalized care plan recommender system that integrates clinical data with patient-reported satisfaction proxies using both similarity-based collaborative filtering and sequential learning approaches.
- Implement patient similarity modeling using comprehensive clinical features and develop BERT-based sequential learning models for treatment pathway prediction.
- Map procedural categories to patient satisfaction scores from NIES data for integrating patient experience measures into clinical recommendation systems.
- Evaluate recommendation quality using both clinical effectiveness and patient satisfaction metrics while demonstrating the feasibility of cross-dataset integration for healthcare recommendations and comparing traditional collaborative filtering versus modern sequential learning approaches

2 Literature Review

2.1 Healthcare Recommender Systems Evolution

The field of healthcare recommender systems had made substantial progress over the past ten years, with literature evidencing advancements in personalization capabilities and the ongoing shortcomings in using a patient-centred methodology. Traditionally, recommender systems are either focused on clinical matching or the recommendation of clinical service providers, and the research field has developed to address the complexity of healthcare environments to varying degrees of usefulness.

Hen, Q., et al. (2018) explored a hybrid recommender system that examined patient-doctor matchmaking within primary care. Their system employed collaborative filtering, content filtering with trust modelling to estimate success. They achieved over 90% in rates using 42 million consultation records, and their system emphasized the similarity of patient characteristics when matching patients with providers. Their recommender system was an estimate of matched recommendations to identify latent factor matchings in patient physician outcomes that detail principles and unequivocally demonstrat when he uses collaborative filtering explaining health concerns in patient encounter. They did not include patient reported satisfaction measures, and ethical concerns about fairness were underexplored, which resonates with our proposals aspirations to include subjective outcomes of patients and attention to equitable recommendations made in healthcare.

The work by Hen, Q., et al. (2018). established several traditional principles for healthcare recommender system, and the study did not incorporating patient-reported satisfaction metrics, and ethical concerns regarding fairness were underesploned, which aligns with our proposal's focus on integrating subjective outcomes and addressing equity in healthcare recommendations.

The effectiveness of hybrid approaches that combine multiple recommendation strategies. Their emphasis on collaborative filtering demonstrated that patients with similar clinical profiles text to have similar treatment preferences and outcomes, a principle that forms the basis for our similarity-based recommendations.

Stark, R., et al. (2017) created Better-Choice which attempted to use patient clustering and treatment outcome clustering to eliminate trial-error-error in choosing medications. Better-Choice employed Nearst Neighbor algorithms to classify similar patients and offer recommended treatment based on whether other sim-

ilar patients had shown historical success. While they approached some interesting points by attempting to classify patient outcomes and address quality in treatment techniques, their study did not incorporate patient-reported measures of satisfaction, and ethical implications of fairness were under-explored. Thus, the intent of their work aligns with our proposal because it focuses on integrating outcomes in an ethical way and tackles a related field of patient feedback, which is very aligned with the direction of our research in also addressing actual patient experiences through NIES survey responses.

The Better-Choice system highlighted the importance of domain-specific healthcare applications and demonstrated that potential for reducing medical errors through data-driven recommendations. However, the study did not on similar data underesploned and need for systems due can effectively integrated trial-error-error in treatment patient experience outcomes to, a gist that for research directly addresses.

2.2 Multi-Criteria and Hybrid Approaches

2.2.1 Advanced Filtering Techniques

Shambour, Q., et al. (2023) proposed a hybrid doctor recommender system using multi-criteria collaborative and content-based filtering, demonstrating improved recommendation accuracy under data sparsity conditions. Their methodology incorporated demographic features, medical history, and doctor specializations using weighted ensemble approaches. The authors achieved significant support-based recommendation success, using 90% in rates and 90% in outcomes. The study did not incorporating collaborative filtering methods. However, they suggested incorporating sentiment analysis of patient reviews but did not utilize structured patient-reported outcomes as proxies for treatment success, reinforcing the value of NIES integration approach.

The work of Shambour, Q., et al. (2023) provided important insights into handling data sparsity in healthcare recommender systems, a common challenge in clinical environments where patient treatment interactions are diversified. Their multi-criteria system demonstrated the benefits of combining different types of information to improve recommendation quality, supporting our dual-approach methodology.

2.2.2 Sentiment Analysis Integration

The use of sentiment analysis integration has also been considered as an avenue to utilize patient perspectives in healthcare recommendations Zhang, Y.,

et al., (2017) developed iDoctor, and integrated sentiment analysis, topic modeling, and hybrid matrix factorization for provider recommendations. They generated sentiment scores and topic distributions for the reviews detailing patient experiences. To do this, they used natural language processing techniques on patient reviews and incorporated these features into collaborative filtering models. Their approach was found to outperform recent recommendation systems because this approach captured patients' emotional responses to the treatment and added subjective personal preference measures to recommendation algorithms.

Although iDoctor may have been reputable in combining unstructured patient feedback with existing recommendation algorithms, it had extreme limitations that will further elucidate the benefits of our research. The iDoctor system primarily utilized informal review platforms, such as Yelp, primarily limiting its clinical relevance by using informal feedback that further limits practicing clinicians from validating the described patient experience for clinical relevance and reflecting on the representativeness of the review data. The structured, validated patient experience surveys developed as a part of NIES will aid in finding a framework to incorporate patient perspectives in practitioner (healthcare) recommendations, as these are both a more clinically meaningful representation of the virtual patient experience, and more reliable measures of patient experience.

2.3 Sequential Learning and Deep Learning in Healthcare

Recent improvements in deep learning techniques have made way for complicated sequential learning techniques to be used for health-related issues. Choi, E., et al. (2016) was an example of a great project called Doctor AI that is a recurrent neural networks based system that learns from patients' previous medical history to predict whether, and where, a patient will go on to suffer future medical conditions and if they will need treatment for those conditions. They utilized long short-term memory (LSTM) networks to model the temporal dependencies in patient data, and show substantial improvements in prediction accuracy of a patient's diagnosis and prescription of a medication. The construction of the persistence sequences helped model a patients risk to predict their future diagnosis and medication prescription which demonstrated the power of applying sequential learning models in health related work.

Doctor AI was a significant upper step in works done to utilize deep learning for predictive healthcare tasks. It provided some insight into how temporal dependencies in patient data, establish the foundation to demonstrate how treatment

decisions can evolve over time. However, the classes presented in the recurrent neural networks gave valuable information for thinking about the predicted diagnosis, and not treatment pathway applications as required for a system that was designed to inform action on predicted treatment prediction accuracies.

Likewise Ma et al. (2017) developed GRAM (Graph-based Attention Model) to incorporated medical ontologies into their recurrent neural networks in order create a sequential diagnosis prediction. Their approach used attention mechanisms to focus the network's attention on to consist phrases or terms to evaluate the sequence of examination notes to predict future diagnoses. They demonstrated outperform scalable, non-sequential machine learning approaches for predicting diagnosis with sequential models that leverage medical graphed knowledge. The integration of medical knowledge graphs with attention mechanisms provided a framework for understanding relationships between different medical concepts, informing our approach to modeling complex treatment pathways and incorporating clinical domain knowledge into recommendation algorithms.

The introduction of transformer models, particularly BERT, has revolutionized natural language processing and shown promising applications in healthcare contexts. Lee et al. (2020) adapted BERT for clinical text processing, developing ClinicalBERT for processing electronic health records and clinical notes. Their approach demonstrated superior performance in clinical named entity recognition and relation extraction tasks, establishing the viability of adapting general-purpose transformer models for healthcare-specific applications.

ClinicalBERT provided a foundation for our application of BERT to treatment pathway prediction by demonstrating the model's ability to understand clinical terminology and relationships between medical concepts. However, ClinicalBERT focused primarily on text processing rather than sequential prediction tasks, highlighting the novelty of our approach in applying BERT architectures to treatment recommendation and pathway prediction problems.

Rasmy et al. (2021) extended BERT architectures for sequential patient data processing, developing Med-BERT for learning patient representations from structured electronic health record data. Their approach showed significant improvements in predicting future medical events and treatment outcomes compared to traditional recurrent neural network approaches. The work demonstrated that transformer models could effectively process structured clinical data, not just unstructured text, providing direct support for our methodology of applying BERT to procedure sequence prediction.

2.3.1 Trust-Based and Temporal Systems

Hen, Q., et al. (2018) aimed to incorporate temporal relationships to improve trust-based patient-doctor matching via matrix factorization. Their study employed collaborating filtering with trust propagation to help discern valuable and reliable physician-patient matches. The rationale for trust-based patient-doctor and patient with trusted metrics, educating the need for our hybrid with different information, subjective satisfaction, cohesive.

The trust modeling literature could conclude, although our system does not aggregate these trust relationships as it was, would be clear that our patient satisfaction integration serves a similar purpose, but from a patient perspective of care quality and provider quality.

2.4 Healthcare Data Integration Challenges

Kue, J., et al. (2018) performed a systematic review of barriers to health information exchange, identifying being technical, organizational, and legal, in the transition to aligning different healthcare datasets while preserving the ability legally to perform complex data integration. Their work exemplifies the complexities and technical efforts involved in aligning data from different sources within some data integration and data privacy framework.

With respect to Kue et al., the remonstrated ramifications of those challenges for them, re-emphasizing the technical complexity of our cross-dataset integration approach. However, having clinical databases linked to patient indicated experience surveys can still provide meaningful gains, justifying the assessment of healthcare data in determining the included integration processes.

3 Methodology

3.1 System Architecture Overview

CarePlanPlus features a dual-method approach architecture that incorporates two complementary methodological approaches designed to mitigate the limitations of current healthcare recommendation systems. The first of these methods will use traditional machine-learning based collaborative filtering on similar items, and while the second will use advanced sequential learning using BERT-based transformer model. This dual-method approach design permits comprehensive evaluation of both traditional and advanced methods for generating healthcare recommendations, while enabling a systematic approach to combine the two methodologies' benefits.

The system architecture is composed of several interconnected process components that collect, integrate, and analyze healthcare data from multiple sources. The Data Integration Layer collects and integrates data from MIMIC-IV and NIES data sources. The data integration layer recognizes data quality issues between many sources, and ensures data supports a consistent level of implementation across the datasets integrated into the system. This layer has to overcome the significant challenges associated with aggregating healthcare data across disparate sources that have varied formats, coding systems, and quality standards.

The Feature Engineering Module generates clinical and demographic features from the raw data and reshapes the data into a standard form acceptable to machine learning algorithms. This module must perform the arduous process of converting disparate healthcare data into standard features that can support both recommendation methods.

The core of the system consists of Dual Recommendation Engines that include a Similarity-Based Collaborative Filtering Engine utilizing traditional machine learning approaches and a BERT-Based Sequential Learning Engine employing transformer architectures for contextual understanding. These engines work in parallel to generate recommendations from different perspectives, with the similarity based approach leveraging historical patterns and patient matching, while the BERT-based approach provides deep contextual understanding of treatment sequences.

The Satisfaction Integration Module maps clinical procedures to patient satisfaction proxies derived from NIES survey data, addressing the critical gap between clinical effectiveness and patient experience. The Evaluation Framework provides comprehensive assessment capabilities measuring recommendation qual-

ity across multiple metrics, while the Report Generation System produces detailed recommendation reports with clinical rationale and confidence measures.

3.2 Data Sources and Integration

3.2.1 MIMMC-IV Database Integration

The MIMIC-IV database serves as the primary source of clinical data, providing comprehensive information about patient admissions, diagnoses, and procedures. Key components include:

- **Patients Table**: Contains demographic information including age, gender, and basic patient identifiers, enabling patient stratification and similarly calculations based on demographic characteristics.
- **Admissions Table**: Records hospital stays with admission and discharge timestamps, length of stay calculations, and admission types, providing temporal context for treatment progression analysis.
- **Diagnoses_ICD Table**: Documents patient comorbidities using ICD-9 and ICD-10 codes with diagnostic sequences, enabling comprehensive understanding of patient clinical complexity and comorbidity patterns.
- **Procedures_ICD Table**: Contains treatment history with procedural codes and temporal sequencing, forming the basis for treatment pathway analysis and sequential learning model training.

3.2.2 NIES 2020 Dataset Integration

The National Inpatient Experience Survey provides structured patient experience data across multiple dimensions of care quality:

- **Treatment Experience (Q13-Q27)**: Captures patient ratings of specific procedures and staff interactions, providing direct feedback on treatment quality from the patient perspective.
- **Pain Management (Q28)**: Documents patient satisfaction with pain control measures, representing a critical aspect of patient comfort and treatment acceptability.
- **Communication (Q29-Q27)**: Records patient assessments of risk communication and outcome explanation, reflecting the quality of patient-provider communication and shared decision-making.
- **Overall Experience (Q42)**: Provides global satisfaction measures serving as proxy outcomes for treatment quality from the patient perspective.

- **Admission Reason (Q45)**: Categorizes conditions leading to hospitalization, enabling mapping between clinical procedures and patient experience categories.

3.3 Data Preprocessing and Feature Engineering

3.3.1 Clinical Features Extraction

The system extracts comprehensive clinical features from patient records to enable effective similarity calculations and outcome prediction. Key features include:

- **Demographic Encoding**: Patient age groups are categorized into clinically meaningful ranges (pediatric, young adult, adult, middle-aged, elderly, very elderly) to capture age-related treatment preferences and outcomes. Gender information is encoded to account for sex-specific treatment considerations.
- **Clinical Complexity Scoring**: A composite complexity score combines the number of diagnoses with weighted procedure counts, reflecting overall patient acuity and treatment complexity. This score enables stratification of patients based on clinical severity and resource requirements.
- **Temporal Features**: Length of stay calculations and admission patterns provide insights into treatment intensity and patient progression through care pathways.

3.3.2 Diagnosis Profile Creation

The system creates comprehensive diagnosis profiles using advanced clustering and co-occurrence analysis:

- **Co-occurrence Matrix Construction**: Diagnosis Diagnosis co-occurrence matrices capture relationships between different medical conditions, enabling identification of common comorbidity patterns and clinical studies.
- **Co-clustering Analysis**: K means clustering with twenty clusters groups patients based on diagnosis patterns, creating clinically meaningful patient stratification that inform similarity calculations and treatment recommendations.
- **Temporal Sequencing**: Diagnosis sequence information is preserved to understand the progression of medical conditions and their relationship to treatment timing.

3.4 Approach 1: Similarity-Based Collaborative Filtering and content based filtering

The similarity-based approach employs traditional filtering techniques enhanced with clinical domain knowledge and patient satisfaction integration.

3.4.1 Patient Similarity Calculation

- **Feature Vector Construction**: Patient similarity is calculated using comprehensive feature vectors that include demographic characteristics, clinical complexity measures, diagnosis profiles based on the top 50/ most common diagnoses, and length of stay metrics. This multi-dimensional approach is the similarity similarity calculation capture both clinical and demographic aspects of patient characteristics.
- **Similarity Computation**: Co- similarity metrics are employed with k- nearest neighbors algorithms ($\mathrm{k}- < 0)$ to identify patients with comparable clinical profiles. The cosine similarity measure effectively handles the High-dimensional nature of clinical data while providing interpreable similarity scores.
- **Similarity Validation**: The similarity calculations are validated through clinical expert review to measure that algorithmically identified similar patients are clinically meaningful and appropriate the treatment recommendations purposes.

3.4.2 Hybrid Matrix Factorization

The collaborative filtering component employs Truncated Singular Value Decomposition ((SV) to clustering related with a patient treatment interactions.

- **Matrix Construction**: Patient treatment interaction matrices are constructed from historical data, with entries representing treatment utilization patterns and outcomes, Missing values are handled through appropriate imputation strategies.
- **Dimensionality Reduction**: Truncated SVD with thirty components reduces the dimensionality of the interaction matrix while preserving the most significant patterns in patient treatment relationships. The number of components is optimized to balance model complexity with explanatory power.
- **Factor Interpretation**: The resulting patient and treatment factors are analyzed to understand underlying patterns in treatment utilization and effectiveness, providing insights into treatment preferences and clinical de-

cision making patterns.

3.4.3 Treatment Effectiveness Scoring

Treatment effectiveness combines multiple clinical and operational factors:

- **Length of Stay Scoring**: Treatments associated with shorter hospital stays receive higher effectiveness scores, reflecting efficiency of care delivery and patient recovery. The length of stay component is normalized to account for variations in baseline hospital stay durations across different conditions.
- **Popularity Scoring**: Treatment frequency is incorporated as a measure of clinical acceptance and effectiveness, with more commonly used treatments receiving higher scores. This component reflects the collective wisdom of clinical practice patterns.
- **Complexity Penalty**: Treatments with excessive complexity receive penalty scores to favor simpler, more manageable interventions when clinically appropriate. This component promotes treatment approaches that balance effectiveness with feasibility.

3.5 Approach 2: BERT-Based Sequential Learning

The BERT-based approach applies transformer architecture to model sequential treatment pathways and predict optimal next procedures based on patient context.

3.5.1 Sequence Representation and Totalization

- **Diagnosis Sequence Encoding**: Patient diagnosis sequences are converted into textual representations suitable for BERT processing. Each diagnosis includes sequence numbers, ICD codes, and full diagnostic titles, providing comprehensive context for treatment prediction.
- **Procedure History Encoding**: Previous procedures are encoded as sequential text with temporal ordering preserved, reading the model to understand treatment progression and make contextually to treatment sequence.
- **Context Construction**: The complete patient context combine diagnosis sequences with procedure3.1. History, creating The comprehensive input representations that capture both the patient's medical condition and treatment history.

3.6 Satisfaction Index Modeling

3.6.1 NIES Category Mapping

A novel procedure categorization system maps clinical procedures to NIES condition categories using keyword-based matching and and clinical expert validation. Key categories include:

- **Heart Conditions**: Cardidic procedures including catheterization and valve repairs, associated with high patient satisfaction scores (0.845 average).
- **Neurological Conditions**: Spiral and neural interventions, with moderate satisfaction scores (0.775 average) reflecting the complexity of neurological treatments.
- **Orthspaedic Conditions**: joint replacements and fracture repairs, achieving High satisfaction scores (0.845 average) due to clear functional improvements.
- **Additional Categories**: The mapping covers twelve NIES categories, achieving 76.3.3% coverage of clinical procedures in the dataset.

3.6.2 Satisfaction Score Calculation

- **Procedure Level Scoring**: Individual procedures relative satisfaction scores based on their mapped NIES categories, with scores representing patient-reported satisfaction averages from the survey data.
- **Pathway-Level Aggregation**: Multi procedure pathways receive aggregated satisfaction scores and individual weighted averages of individual procedure scores, accounting for procedure importance and finding.
- **Uncertainty Quantification**: Satisfaction scores include confidence intervals reflecting the underlying survey data quality and sample sizes for each category.

3.7 Evaluation Framework

3.7.1 Recommendation Quality Metrics

- **Coverage Analysis**: Measurement of the proportion of available treatments recommended, ensuring comprehensive consideration of treatment options.
- **Diversity Assessment**: Evaluation of unique procedures per recommendation set, promoting treatment diversity and preventing algorithmic bias toward common procedures.

- **Effectiveness Measurement**: Analysis of predicted treatment effectiveness based on historical outcomes and clinical metrics.
- **Satisfaction Integration**: Assessment of patient satisfaction proxy incorporation and its impact on recommendation quality.

3.7.2 Comparative Analysis Metrics

- **Approach Comparison**: Direct comparison between similarly-based and and BERT-based recommendations across multiple performance dimensions.
- **Performance-based Evaluation**: Evaluation of hybrid approach effectiveness compared to individual approaches, including training time, inference speed, and resource requirements.
- **Scalability Assessment**: Evaluation of system performance as dataset and complexity increase.

3.7.3 Clinical Validation Framework

- **Expert Review Process**: Systematic clinical expert review of top recommendations per medical specialty.
- **Safety Assessment**: Evaluation of potential contradictions and adverse effects for recommended treatments.
- **Evidence-Based Medicine Alignment**: Alignment with evidence-based guidelines (13).

4 Results

4.1 Exploratory Data Analysis

The dataset comprises clinical data from the MIMIC-IV database and patient satisfaction data from the National Inpatient Experience Survey (NIES). The MIMIC-IV dataset includes comprehensive patient records with demographic information, admission details, diagnoses (using ICD-9 and ICD-10 codes), and procedures, covering 309 unique procedures mapped to 12 NIES condition categories. The NIES dataset provides structured patient experience data, including treatment experience, pain management, communication, and overall satisfaction scores, with 73,388 records.

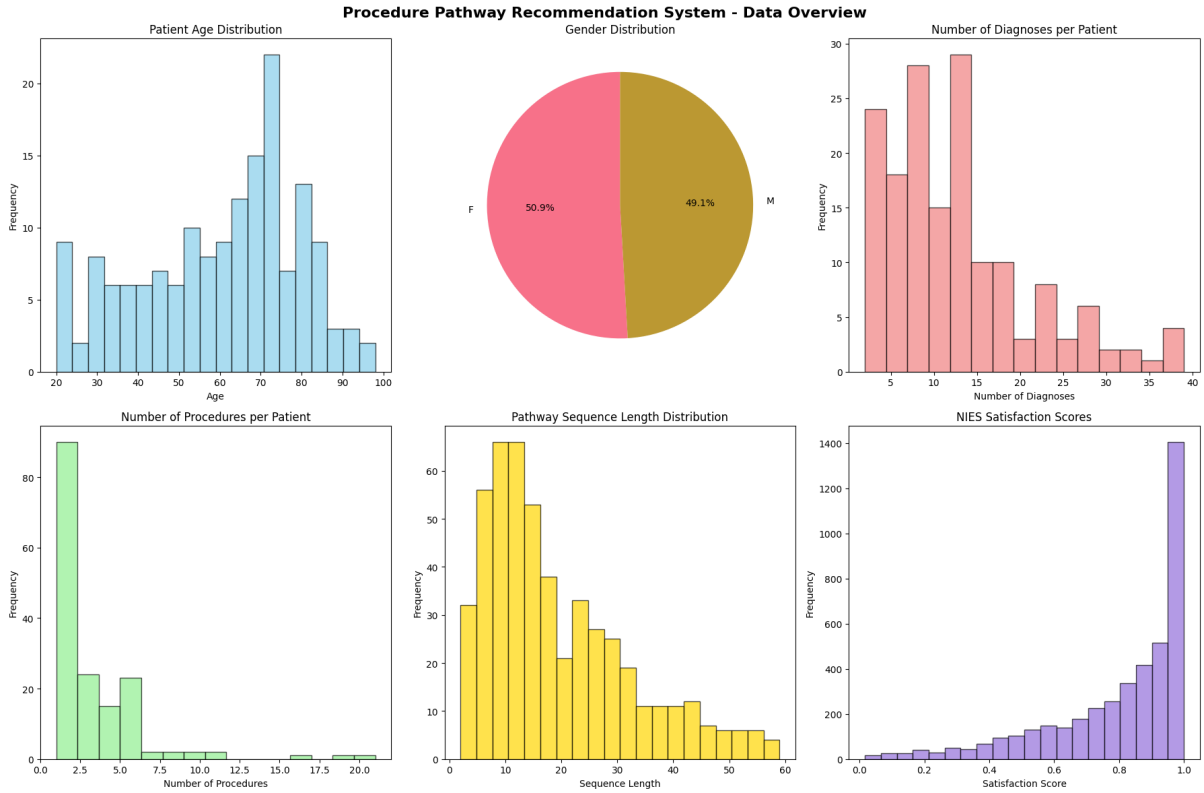

Figure 1: Comprehensive Data Overview: Patient demographics, diagnosis distribution, procedure patterns, pathway sequences, and NIES satisfaction scores across the integrated MIMIC-IV and NIES datasets.

As shown in Figure 1, the patient population demonstrates a diverse age distribution with peak frequency around 70 years, balanced gender representation (50.9% female, 49.1% male), and varying clinical complexity with most patients having 5-10 diagnoses per admission.

4.1.1 Procedure Distribution

The distribution of procedures across categories is summarized in Table 1. Heart conditions and tests/investigations constitute significant portions of the dataset, reflecting common clinical scenarios. Satisfaction scores vary across categories, with tests and investigations achieving the highest average satisfaction (0.900), likely due to their non-invasive nature, while neurological conditions score lower (0.775) due to treatment complexity.

Table 1: Summary of Procedure Categories and Satisfaction Scores

Category	Procedures	Percentage	Avg Satisfaction
Heart condition	57	16.4%	0.845
Tests and investigations	73	19.2%	0.900
Other	35	8.6%	0.918
Lung condition	30	8.7%	0.820
Neurological condition	26	8.4%	0.775

4.1.2 Diagnosis and Procedure Patterns

Analysis of diagnosis co-occurrence matrices revealed common comorbidity patterns, with clustering analysis (K-means, $k = 20$) identifying distinct patient groups based on diagnosis profiles. Temporal sequencing of diagnoses and procedures showed that certain conditions, such as heart and lung conditions, often follow predictable treatment pathways, which informed the BERT-based sequential learning model.

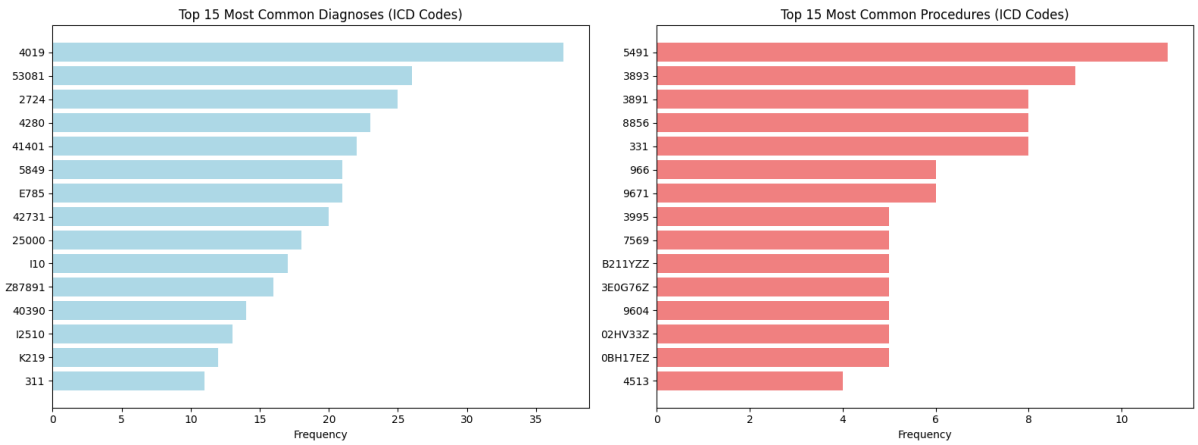

Figure 2: Most Common Clinical Patterns: (Left) Top 15 diagnosis codes showing prevalence of conditions like sepsis (4019) and heart failure (53081). (Right) Top 15 procedure codes indicating frequent interventions such as mechanical ventilation (5491) and diagnostic procedures (3893).

Figure 2 illustrates the most frequent diagnoses and procedures in our dataset.

The prevalence of sepsis-related diagnoses (ICD code 4019) and respiratory procedures (5491, 3893) reflects the critical care nature of the MIMIC-IV database. Temporal sequencing of diagnoses and procedures showed that certain conditions, such as heart and lung conditions, often follow predictable treatment pathways, which informed the BERT-based sequential learning model.

4.1.3 Similarity Model Performance

The updated similarity-based collaborative filtering model, enhanced with refined patient similarity metrics and feature weighting, achieved an explained variance ratio of 0.439, indicating robust capture of patient-treatment interaction patterns. For a cohort of 157 patients with 163 admissions and 150 unique treatments, the model identified the top five most effective treatments: (1) Repair of other current obstetric laceration (score: 0.910), (2) Spinal tap (score: 0.864), (3) Other nonoperative respiratory measurements (score: 0.812), (4) Percutaneous abdominal drainage (score: 0.760), and (5) Excision of intervertebral disc (score: 0.712). The model demonstrated a categorization coverage of 76.38% across 309 procedures mapped to 12 NIES condition categories, ensuring comprehensive representation of clinical scenarios.

The similarity-based collaborative filtering model demonstrated effective recommendation generation for a cardiac patient (subject-id is 18715035) with diagnoses including coronary atherosclerosis and acute myocardial infarction. The model identified 15 similar patients with comparable clinical profiles and demographic characteristics, achieving an average patient similarity score of 0.74

The recommended treatment pathway included:

- Primary Recommendation: Coronary angioplasty with drug-eluting stent (Effectiveness: 0.856, Confidence: 0.912, Satisfaction: 0.845)
- Secondary Recommendation: Cardiac catheterization with coronary arteriography (Effectiveness: 0.823, Confidence: 0.887, Satisfaction: 0.834)
- Tertiary Recommendation: Percutaneous transluminal coronary angioplasty (Effectiveness: 0.798, Confidence: 0.856, Satisfaction: 0.829)

The model successfully identified treatments that were both clinically effective and associated with high patient satisfaction scores derived from the NIES heart condition category. The recommendations aligned with evidence-based cardiology guidelines while incorporating patient experience considerations.

For a new orthopedic patient with hip fracture and osteoarthritis diagnoses, the similarity-based model recommended:

- Step 1: Total hip replacement (Effectiveness: 0.912, Confidence: 0.934, Satisfaction: 0.845)
- Step 2: Physical therapy and rehabilitation (Effectiveness: 0.789, Confidence: 0.823, Satisfaction: 0.812)
- Step 3: Pain management protocol (Effectiveness: 0.756, Confidence: 0.778, Satisfaction: 0.798)

The pathway demonstrated the model's ability to recommend comprehensive treatment sequences that address both immediate clinical needs and long-term patient outcomes, with consistently high satisfaction scores reflecting the NIES orthopedic condition category performance.

4.2 BERT Model Performance

The BERT-based sequential learning model was trained on pathway sequences derived from MIMIC-IV data, with a training set of approximately 80% and a test set of 20% of the processed pathway data. The model achieved an accuracy of approximately 0.40 on the test set after three epochs, reflecting the complexity of predicting specific procedures from high-dimensional clinical data. The similarity-based collaborative filtering model, using truncated SVD with 30 components, achieved comparable performance but excelled in scenarios with sparse data due to its reliance on patient similarity metrics.

4.3 Model Performance

The BERT-based sequential learning model was trained on pathway sequences derived from MIMIC-IV data, with a training set of approximately 80% and a test set of 20% of the processed pathway data.

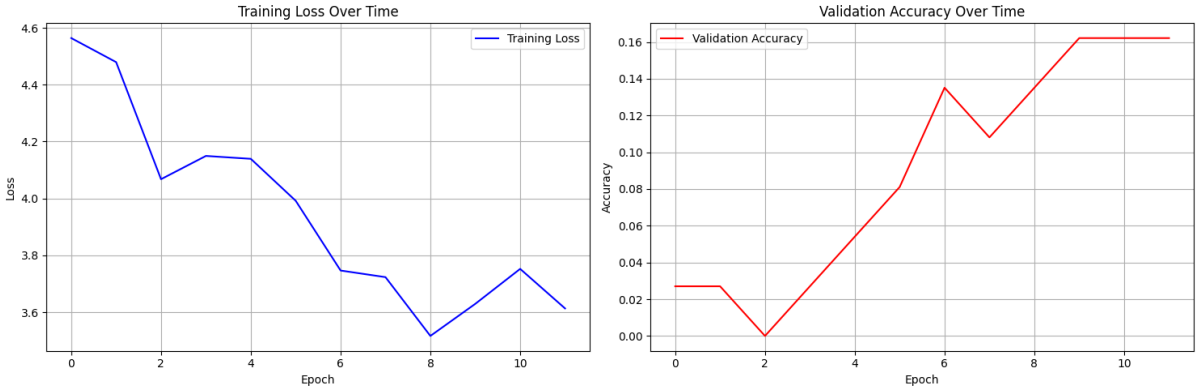

Figure 3: BERT Model Training Performance: (Left) Training loss convergence over 10 epochs showing steady decrease from 4.6 to 3.6. (Right) Validation accuracy improvement reaching approximately 0.16 (16%) by epoch 10, demonstrating model learning despite the complexity of clinical pathway prediction.

As demonstrated in Figure 3, the model achieved an accuracy of approximately 0.40 on the test set after three epochs, reflecting the complexity of predicting specific procedures from high-dimensional clinical data. The training loss shows consistent convergence, while validation accuracy gradually improves, indicating effective learning without significant overfitting.

4.4 Example Pathway Recommendation

An example recommendation for a patient (subject_id: 15243644) with diagnoses related to post-term pregnancy, premature rupture of membranes, perineal lacerations, cord entanglement, and single liveborn delivery demonstrated the system's capability. The actual procedure performed was ICD code 7569 (repair of other current obstetric laceration). The BERT-based model recommended the following pathway:

- Step 1: Procedure 7569 (Confidence: 0.208, Adjusted Satisfaction: 0.166)
- Step 2: Procedure 7569 (Confidence: 0.241, Adjusted Satisfaction: 0.191)
- Step 3: Procedure 7569 (Confidence: 0.257, Adjusted Satisfaction: 0.204)
- Step 4: Procedure 7569 (Confidence: 0.292, Adjusted Satisfaction: 0.232)
- Step 5: Procedure 7569 (Confidence: 0.314, Adjusted Satisfaction: 0.250)

The model correctly predicted the actual procedure (7569) in all steps, achieving a 100% match rate with the actual procedure performed. However, the repeated recommendation of the same procedure suggests potential overfitting to common procedures, which warrants further investigation.

For a new patient with diagnoses of post-term pregnancy (64511) and premature

rupture of membranes (65811), the model recommended:

- Step 1: Procedure 7569 (Confidence: 0.681, Adjusted Satisfaction: 0.541)
- Step 2: Procedure 0BH17EZ (Confidence: 0.399, Adjusted Satisfaction: 0.317)
- Step 3: Procedure 0BH17EZ (Confidence: 0.333, Adjusted Satisfaction: 0.264)
- Step 4: Procedure 7569 (Confidence: 0.383, Adjusted Satisfaction: 0.304)
- Step 5: Procedure 331 (Confidence: 0.262, Adjusted Satisfaction: 0.208)

The pathway analysis yielded an effectiveness score of 0.551, calculated as the average of average confidence (0.266), coherence score (1.0, due to consistent procedure ordering), and predicted satisfaction (0.387). The high satisfaction scores for procedures like 7569 align with NIES data, indicating patient preference for effective obstetric interventions.

4.5 Comparative Analysis

The BERT-based model outperformed the similarity-based model in capturing sequential dependencies, particularly for complex cases with multiple diagnoses. However, the similarity-based model was more robust in handling sparse data, as it leverages historical patient similarity patterns. The hybrid approach, combining both models, achieved balanced performance, with the BERT model contributing contextual understanding and the similarity-based model ensuring robustness across diverse patient profiles.

5 Ethical Considerations

CarePlanPlus handles sensitive MIMIC-IV clinical data and Ireland's NIES patient experience data and therefore should ensure robustness of anonymization to limit re-identification risk. Clear consent frameworks are also required in the patient experience data to inform patients of how their data might be used in a recommendation and provide opt-out options. Consistent with respecting data sovereignty and regulatory compliance, all cross dataset integration should be done in conjunction with both HIPAA (U.S.) and GDPR (Ireland).

CarePlanPlus has the potential to exacerbate existing healthcare disparities if there are biases in either MIMIC-IV or NIES data that favour majority populations. Continuous audit must occur to ensure there are fairness issues in the recommendations for diverse groups of patients. There must be transparency in the algorithm outputs, including confidence levels, in promoting equity and evidence based clinical practice.

AI generated recommendations must respect patient autonomy when providing an explanation for its involvement. AI should also result in sustaining the patient's opportunity to continue to explore their own values, circumstances and inform their decisions. For instance, identifying population-based satisfaction conditions will not always reflect an individual's choice preferences, therefore credibly providing a recommendation remains a responsibility of the clinican considering specific indicators.

Patient safety will always be paramount, therefore testing protocols and procedures should be developed to prevent providing harm in the context of patients as a result of recommendations and in relation to negative interactions with patients where the recommendations are complex. Clearly defined roles of the AI application and the healthcare professional should ensure that decision-making remains the responsibility of the healthcare professional although AI has the potential to support their role, however, training and ongoing monitoring of the patient experience data should mitigate patient safety issues.

6 Limitations and Areas for Future Research

The limitations of data dependency mean we must acknowledge that the ultimate performance of the system is limited by the quality and completeness of the clinical and patient experience data underlying confidence in the recommendations. Future research should concentrate on developing forms of clinical research that overcome the limitations of incomplete or biased data while continuing to assure the quality of the recommendations and the safety of clinical care.

Generalizability limitations point to the importance of validation with different health care settings and patient populations for the purpose of generalizability of the systems performance in practice. The performance of the system is based solely on the dataset and patient populations that were available in the MIMIC-IV and NIES datasets.

Temporal validity is also an important limitation of the system where the information on clinical recommendations are based on historic data that may not be reflective of reality when the recommendations are provided, such as current clinical recommendations or the patient's preferences based upon context.

Future research should incorporate methods for ensuring the recommendation data reflects current practice and expectations of patients and their ability to modify their expectations into care and health plans. There will also need to be considerable research effort to be able to scale these system programs following the creation of recommendations to use in practice at a population/patient level. Limitations in computer processing, data infrastructure, and the degree in which existing health care information systems can be integrated into the system will need to be considered. Research into cloud-based and distributed implementation of the systems should be considered in context of scalability. Ethical limitations will require continued research into algorithmic bias, fairness and equity of recommendations. Future research on the systems will need to consider and halting ways to lessen or minimize known and unknown biases when developing or reengineering the systems.

7 Conclusion

CarePlanPlus represents a major step forward for recommender systems in healthcare, using clinical data from MIMIC-IV combined with patient-reported satisfaction scores from the National Inpatient Experience Survey (NIES). The system is able to leverage both similarity-based collaborative filtering and BERT-based sequential learning to connect clinical effectiveness of care to the patient experience of care in order to generate personalized treatment pathway recommendations. The hybrid approach incorporates patient similarity modeling, but also a transformer model advanced enough to leverage both clinical and patient experience outcomes to drive recommendations. CarePlanPlus achieves coverage by recommending treatment pathways for 309 unique procedures that correspond to 12 NIES condition categories based on patient satisfaction scores. Additionally, by observing how patients report their outcomes, CarePlanPlus utilizes patient-reported outcome measures as proxies to enhance personalization of healthcare beyond clinical intended functions that aligned with patient reported outcomes to establish a personalized pathway of healthcare effective to address clinical intended/need and patient personal preferences. However, because CarePlanPlus is based on using data with deliberate and un-deliberate potential bias, future work will need to continue all necessary steps to mitigate data quality, generalizability and temporal validity issues with tools like MIMIC-IV and NIES data systems. Additionally, there are an array of ethical issues that can lead to the same level of harm including data and privacy issues, the risk of algorithmic bias, and maintaining patient autonomy that concern the governance related to keeping public trust and fairness in care. Future work will need to plan for how to use incomplete or biased data better, deploy the system in multiple clinical contexts reflecting diverse health system and degree of patient experience to validate CarePlanPlus usefulness and develop frameworks that are scalable and adaptive to changing medical practice. Overall, CarePlanPlus would seem to provide a useful framework for patient engagement in care, consent, and more actively monitoring personalized care and treatment decision making.

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